established a studio in Pune. It was the ‘biggest transformation journey; it impacted methods, processes, and most importantly, the way individuals worked’, says Umamaheshwaran.

And to ensure that the engineers became familiar with the best drawing and design tools in the world, Ratan sent them to famous international locations to study both technology and products. When he himself was travelling abroad, he ‘would get critical information’ and connect the technical wing with the world experts. In the process, he ensured that the TELCO R&D division was completely revamped and modernized, adds Umamaheshwaran.

‘[The] engineering capability of Tata Motors is because of him [Ratan]. Absolutely,’ says Wagh. The ‘design and engineering capability today in Tata Motors is only because of him’, and he ‘continuously invested in it’. He remarks that Ratan created a crash lab in India in the 1990s, before any other automobile manufacturer in the country. Ratan also went on to create emission labs and established the Tata Motors European Technical Centre: steps necessary to upgrade and benchmark its products to global standards. The only Indian car company comparable with large car makers like Hyundai, etc. was Tata Motors.

Ratan was intimately involved in the formulation of various TELCO designs; he ‘had a feel for design’, Wagh says. He even established a mini review studio in his office at Bombay House. It was a high-end SiliconGraphics (SGI) workstation where Ratan would review the design proposals that were sent to him. He gave his comments and reviews on designs in real time, leaving his mark on virtually every change.

In 1995, Ratan established the Tata AutoComp (TACO) Systems as a specialized agency to provide automotive products and services, primarily for the new products that TELCO was planning. It played a significant role in the rollout of new models. Today, TACO has grown into a large enterprise with fifty-one manufacturing facilities spread across India, North America, Latin America, Europe and China, and nine JVs.

A Holistic Plan

Four years into his tenure as chairman, Tata Sons, Ratan revealed the plans he had been implementing for TELCO and their expected outcomes. As chairman of both TISCO and TELCO, he pushed them to adopt two